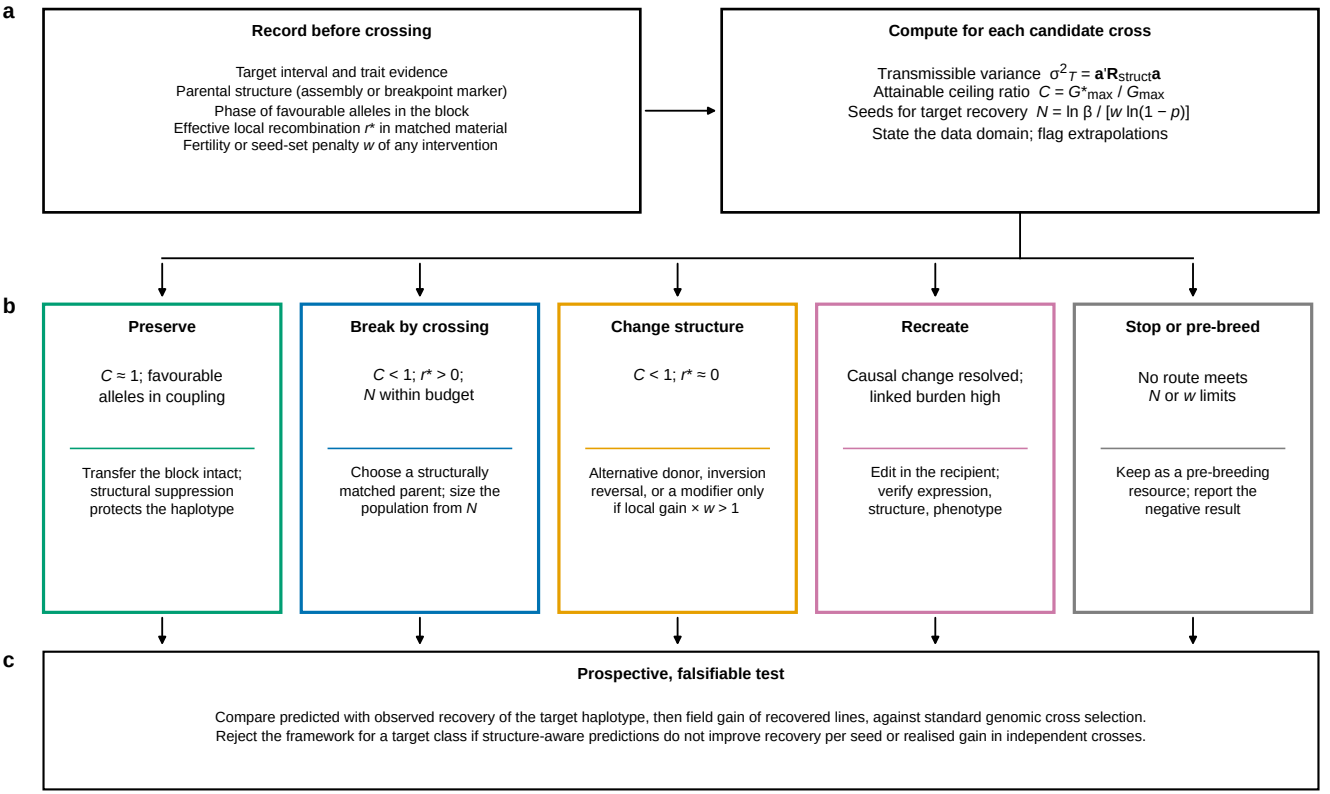
a
Record before crossing
Target interval and trait evidence
Parental structure (assembly or breakpoint marker)
Phase of favourable alleles in the block
Effective local recombination r* in matched material
Fertility or seed-set penalty w of any intervention
Compute for each candidate cross
Transmissible variance σ²T = a'RstructA
Attainable ceiling ratio C = G*max / Gmax
Seeds for target recovery N = ln β / [w ln(1 − p)]
State the data domain; flag extrapolations
b
Preserve
C ≈ 1; favourable alleles in coupling
Transfer the block intact; structural suppression protects the haplotype
Break by crossing
C < 1; r* > 0; N within budget
Choose a structurally matched parent; size the population from N
Change structure
C < 1; r* ≈ 0
Alternative donor, inversion reversal, or a modifier only if local gain × w > 1
Recreate
Causal change resolved; linked burden high
Edit in the recipient; verify expression, structure, phenotype
Stop or pre-breed
No route meets N or w limits
Keep as a pre-breeding resource; report the negative result
c
Prospective, falsifiable test
Compare predicted with observed recovery of the target haplotype, then field gain of recovered lines, against standard genomic cross selection.
Reject the framework for a target class if structure-aware predictions do not improve recovery per seed or realised gain in independent crosses.